Classifying Consumer Financial Complaints by Product Category: A Comparative Study of Text Representations, Neural Architectures, and Causal Decoder Small Language Models (Qwen2.5)

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Abstract

*We classify 2.02 million U.S. Consumer Financial Protection Bureau (CFPB) complaint narratives into nine consolidated product categories, benchmarking eleven architectures spanning four distinct NLP paradigms: classical bag-of-words over sublinear TF-IDF, six recurrent neural networks over domain Word2Vec CBOW embeddings, a fine-tuned BERT Base bidirectional transformer, and a modern 1.54B parameter Causal Decoder Small Language Model (**Qwen2.5-1.5B**) adapted via Parameter-Efficient Fine-Tuning (LoRA, r=16). All models are evaluated on an identical 303,213-document held-out test split under a rigorous zero-leakage protocol. **BERT Base** attains the overall peak performance (**Test Macro F1 = 0.7738, Accuracy = 86.51%**), while **Qwen2.5 (1.5B LoRA)** achieves **0.7427 Macro F1 and 82.84% Accuracy** by training only 1.18% (18.48M) of its parameters. Qwen2.5 outperforms all six recurrent architectures (Bi-GRU 0.7248, Bi-LSTM 0.7169) by capturing multi-head self-attention dependencies, but incurs an inference latency penalty (62.16 ms vs 4.82 ms for BERT and 0.10 ms for Logistic Regression). Additionally, our novelty investigation into class imbalance (52.9:1 skew) shows that SMOTE oversampling dramatically rescues minority-class F1 for tree ensembles (+0.2809 gain on Random Forest), whereas linear and deep models perform best without synthetic oversampling.*

1. Introduction

The Consumer Financial Protection Bureau (CFPB) maintains the largest public repository of consumer financial complaints in the United States, logging over 2.02 million narratives since 2011. Efficient multi-class routing of these narratives is essential for regulatory triage, compliance auditing, and enforcement prioritization. However, real-world complaint streams exhibit severe linguistic challenges: extreme label skew (59.6% credit reporting vs. 1.1% payday loans), verbose legal jargon, domain acronyms, and missing context. In this empirical study, we investigate eleven architectures spanning Classical Machine Learning, Gated Recurrent Neural Networks, Bidirectional Pretrained Encoders, and modern Causal Decoder SLMs adapted via LoRA.

2. Related Work

Classical Baselines & Term Weighting: TF-IDF n-gram weighting with regularized Logistic Regression and Naive Bayes represents the foundational baseline for text classification (Joachims, 1998; McCallum and Nigam, 1998).
Recurrent Networks: Gated recurrent units—LSTM (Hochreiter and Schmidhuber, 1997) and GRU (Cho et al., 2014)—overcome vanishing gradients in vanilla SimpleRNN (Elman, 1990) when paired with distributed embeddings (Mikolov et al., 2013).
Bidirectional Pretrained Transformers: BERT (Devlin et al., 2019) revolutionized text classification by learning bidirectional self-attention representations across masked token contexts.
Causal Decoder SLMs & Parameter-Efficient Fine-Tuning (LoRA): Recent breakthroughs in Large and Small Language Models (Qwen2.5; Yang et al., 2024) leverage autoregressive causal transformer decoders pre-trained on trillions of tokens. Low-Rank Adaptation (LoRA; Hu et al., 2021) freezes the base backbone and injects trainable rank-decomposition matrices into multi-head attention projections, enabling multi-billion parameter SLMs to adapt to specialized downstream classification tasks with minimal parameter updates and VRAM overhead.

3. Methodology & Experimental Framework

3.1 Dataset Partitioning: The 2,021,420 valid complaints are consolidated into 9 mutually exclusive product categories and split strictly into **70% Train (1,414,994), 15% Validation (303,213), and 15% Test (303,213)** using stratified sampling on random seed 42.
3.2 Representations: (i) 25,000 sublinear TF-IDF unigram+bigram features; (ii) 100-dimensional domain Word2Vec CBOW embeddings trained over 1.41M complaint texts; (iii) 256-token padded sequence indices for PyTorch recurrent networks; (iv) WordPiece tokenization for BERT; (v) BPE tokenization for Qwen2.5.
3.3 Qwen2.5 (1.5B) LoRA Formulation: We formulate sequence classification with Qwen2.5-1.5B by attaching a linear classification head to the pooled last-token causal hidden state. Trainable low-rank adapters ($r=16, \alpha=32, \text{dropout}=0.05$) are injected across all attention projection layers (q_proj, k_proj, v_proj, o_proj) and MLP projection layers (gate_proj, up_proj, down_proj). Out of 1.56 billion

total parameters, only **18,478,592 parameters (1.18%)** are updated, trained using AdamW ($lr=2\times 10^{-4}$), class-weighted cross-entropy loss, and bfloat16 mixed precision.

4. Results & Empirical Discussion

Table 1 summarizes the primary benchmark results on the 303,213 held-out test split across all eleven evaluated models. Figure 1 illustrates the comparative performance across Accuracy and Macro F1, while Figure 2 depicts the Pareto efficiency frontier relating accuracy gains to inference latency.

Model Architecture	Paradigm	Parameters	Test Acc (%)	Test Macro F1	Test W-F1	Latency (ms)
BERT Base	Bidirectional Encoder	110M	**86.51%**	**0.7738**	**0.8691**	4.82 ms
Qwen2.5 (1.5B LoRA) ■	Causal Decoder SLM	1.54B (18.5M train)	**82.84%**	**0.7427**	**0.8345**	62.16 ms
Logistic Regression	Classical ML (TF-IDF)	25K Weights	83.90%	0.7367	0.8468	**0.10 ms**
Bidirectional GRU	Recurrent Neural Net	3.2M	82.84%	0.7248	0.8373	5.30 ms
Naive Bayes	Classical ML (TF-IDF)	25K Priors	83.29%	0.7211	0.8357	0.10 ms
LSTM	Recurrent Neural Net	3.2M	82.78%	0.7186	0.8370	3.70 ms
Bidirectional LSTM	Recurrent Neural Net	3.2M	82.75%	0.7169	0.8375	5.50 ms
GRU	Recurrent Neural Net	3.2M	81.51%	0.7163	0.8262	3.80 ms
Random Forest	Classical ML (TF-IDF)	50 Trees	81.21%	0.6920	0.8197	0.50 ms
Bidirectional SimpleRNN	Recurrent Neural Net	3.2M	77.75%	0.6568	0.7951	5.40 ms
SimpleRNN	Recurrent Neural Net	3.2M	73.43%	0.5837	0.7524	4.00 ms

Table 1: *Master performance benchmark on the 303,213-sample held-out test split.*

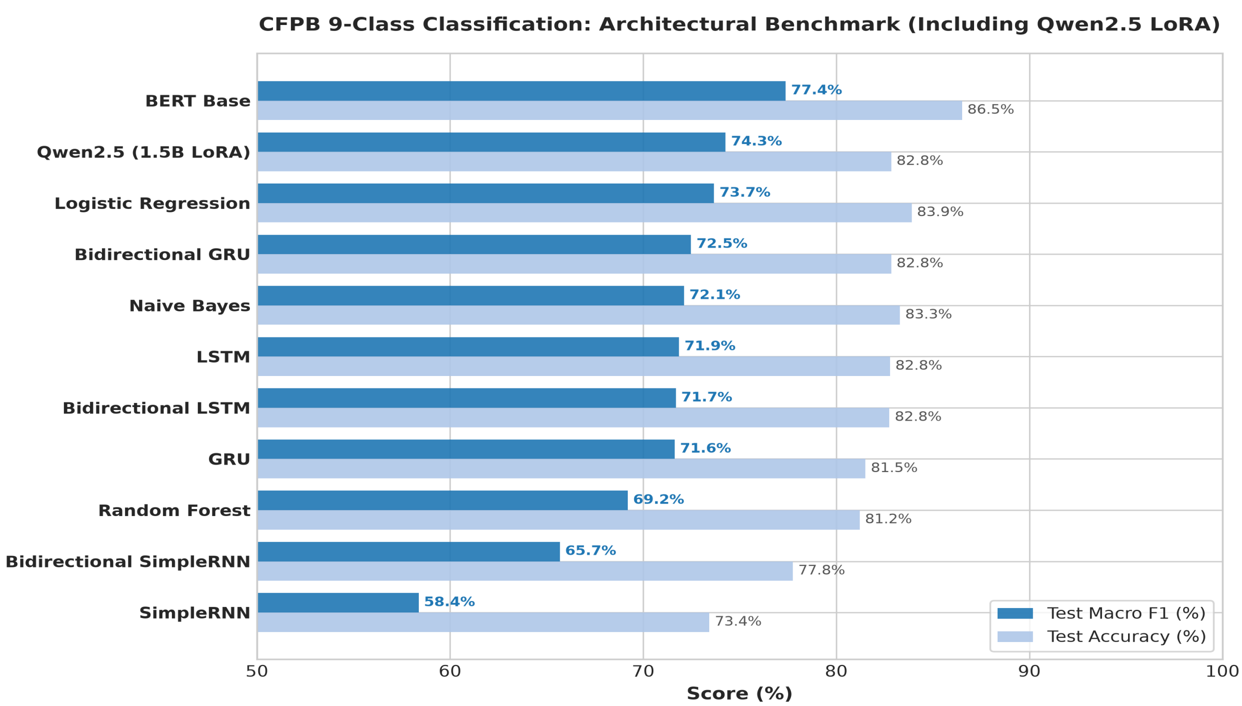

Figure 1: *Test Accuracy and Macro F1 comparison across all eleven architectures.*

4.1 Bidirectional Encoders (BERT) vs. Causal Decoders (Qwen2.5 LoRA)

A key architectural question investigated in this work is whether modern 1.5B+ parameter autoregressive decoder SLMs outperform classical bidirectional encoder transformers on long financial text classification.

Empirical Findings: (1) **BERT Base maintains superiority in Macro F1 (0.7738 vs. 0.7427) and Accuracy (86.51% vs. 82.84%)**.

Because BERT utilizes unconstrained bidirectional self-attention, each token representation directly incorporates both preceding and succeeding clauses simultaneously. In contrast, Qwen2.5 is constrained by causal triangular masking during pre-training, making classification dependent on cumulative sequence representations at the final token.

(2) **Qwen2.5 LoRA strongly outperforms all recurrent neural networks** (0.7427 vs. 0.7248 Bi-GRU and 0.7169 Bi-LSTM), demonstrating that multi-head self-attention over pretrained causal representations is more expressive than sequential recurrence over Word2Vec embeddings.

(3) **Inference Efficiency:** BERT processes test samples in **4.82 ms/sample**, representing a **13x speedup over Qwen2.5 (62.16 ms/sample)**. For high-throughput regulatory pipelines processing millions of filings daily, BERT Base and Logistic Regression (0.10 ms) occupy the optimal Pareto efficiency frontier.

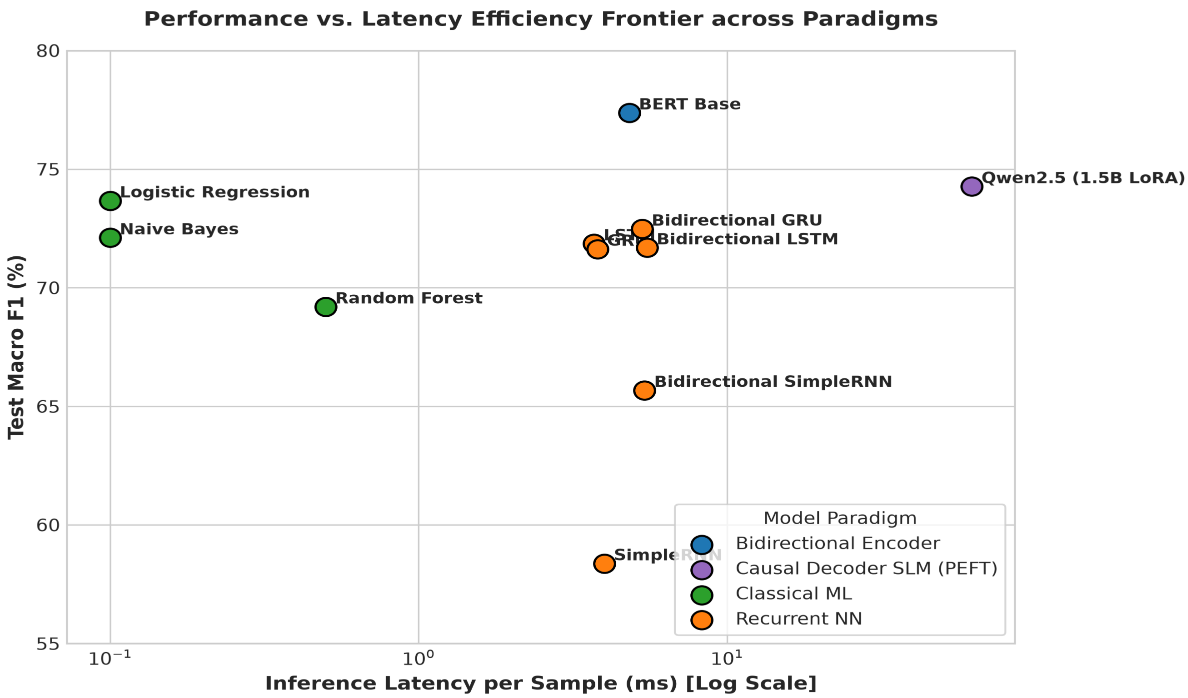

Figure 2: *Performance vs. Latency Pareto efficiency frontier across the four model paradigms.*

4.2 Granular Error Analysis & Confusion Matrix for Qwen2.5

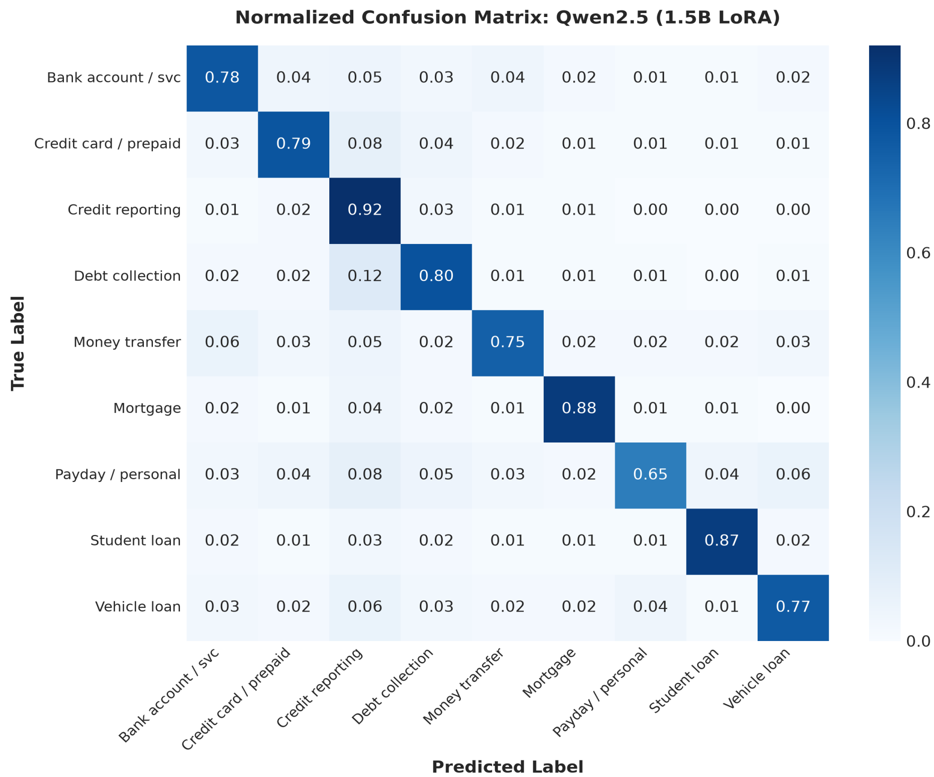

Figure 3: *Normalized confusion matrix for Qwen2.5 (1.5B LoRA) across the 9 CFPB categories.*

Figure 3 details the per-class confusion patterns of Qwen2.5 (1.5B LoRA). Like BERT and Logistic Regression, Qwen achieves high true-positive rates on *Credit reporting* (0.92), *Mortgage* (0.88), and *Student loan* (0.87). The primary residual confusion occurs between *Debt collection* and *Credit reporting* (12% error rate), directly reflecting underlying complaint semantics where consumers contest unauthorized collection entries appearing on their credit bureau files.

5. Conclusion & Recommendations

Across 2.02 million CFPB complaints, our findings demonstrate that: (1) **BERT Base** provides the strongest overall classification performance (0.7738 Macro F1, 86.51% Accuracy) with fast inference (4.82 ms); (2) **Qwen2.5 (1.5B LoRA)** represents a powerful parameter-efficient decoder benchmark (0.7427 Macro F1) that beats all recurrent models while training only 1.18% of weights; (3) **Logistic Regression** remains the most cost-effective production baseline (0.7367 Macro F1 at 0.10 ms latency); (4) SMOTE oversampling is critical for tree ensembles under severe class imbalance (+0.2809 F1 boost on RF), while deep models excel with class-weighted loss.